

QMux

draft-ietf-quic-qmux-02
QUIC – IETF 126 – 2026-07

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Since IETF 125

- Published 01 and 02 drafts
- 00 -> 01 wire breaking changes
- 01 -> 02 no wire breaking changes
- Key changes (discussed previously)
 - QMux records - [Section 3.2](#)
 - QMux over TLS must use ALPN - [Section 8.1](#)
 - Allow use of RESET_STREAM_AT - [Section 9.2](#)
 - General hardening, cleanup, clarification

Implementations of draft 01+

- quicly
- HAproxy
- mvfst
- picoquic
- qmux msquic fork
- Cloudflare quiche fork
 - Curl backend for HTTP/3 over QMux
- quic-go fork
- moq-dev/web-transport/qmux
- @moq/qmux
- Wireshark Dissector MR

Interop of draft 01+

- What is a good interop driver?
- QUIC interop traditionally used HTTP
 - [draft-kazuho-httpbis-http3-over-qmux-00](#)
 - *"h3qx-01", which represents HTTP/3 over draft-ietf-quic-qmux-01.*
 - draft-ietf-quic-qmux-02 is wire-compatible, no need for another ALPN right now
- MoQ over QMux is also popular target
- There has been interop success, capturing the results has not been so great
- Marco Munizaga has WIP QUIC interop runner [proposal PR](#)
- Implementations doing more than just QMux over TLS
 - Unix sockets, WebSockets, ...

No.	Time	Source	Src Port	Destination	Dst Port	Protocol	Length	fubar	Info
1	0.000000000	127.0.0.1	49860	127.0.0.1	4433	TCP	74		49860 → 4433 [SYN] Seq=0 Win=6
2	0.000018608	127.0.0.1	4433	127.0.0.1	49860	TCP	74		4433 → 49860 [SYN, ACK] Seq=0
3	0.000041696	127.0.0.1	49860	127.0.0.1	4433	TCP	66		49860 → 4433 [ACK] Seq=1 Ack=1
4	0.000459780	127.0.0.1	49860	127.0.0.1	4433	TLSv1.3	583		Client Hello (SNI=localhost)
5	0.000473296	127.0.0.1	4433	127.0.0.1	49860	TCP	66		4433 → 49860 [ACK] Seq=1 Ack=5
6	0.003373793	127.0.0.1	4433	127.0.0.1	49860	TLSv1.3	1307		Server Hello, Change Cipher Sp
7	0.003398753	127.0.0.1	49860	127.0.0.1	4433	TCP	66		49860 → 4433 [ACK] Seq=518 Ack
8	0.004674803	127.0.0.1	49860	127.0.0.1	4433	TLSv1.3	130		Change Cipher Spec, Finished
9	0.006101947	127.0.0.1	4433	127.0.0.1	49860	TLSv1.3	565		New Session Ticket, New Sessio
10	0.006119489	127.0.0.1	49860	127.0.0.1	4433	TLSv1.3	147		Application Data
11	0.006444846	127.0.0.1	4433	127.0.0.1	49860	TLSv1.3	157		Application Data
12	0.006681138	127.0.0.1	49860	127.0.0.1	4433	TLSv1.3	256		Application Data
13	0.007300149	127.0.0.1	4433	127.0.0.1	49860	TLSv1.3	164		Application Data
14	0.007704646	127.0.0.1	49860	127.0.0.1	4433	TLSv1.3	96		Application Data
15	0.007989843	127.0.0.1	49860	127.0.0.1	4433	TCP	66		49860 → 4433 [FIN, ACK] Seq=86
16	0.008189303	127.0.0.1	4433	127.0.0.1	49860	TCP	66		4433 → 49860 [FIN, ACK] Seq=19

Type: application_layer_protocol_negotiation (16)

Length: 10

ALPN Extension Length: 8

▼ ALPN Protocol

ALPN string length: 7

ALPN Next Protocol: h3qx-01

▼ Handshake Protocol: Certificate

Handshake Type: Certificate (11)

Length: 762

Certificate Request Context Length: 0

Certificates Length: 758

► Certificates (758 bytes)

▼ Handshake Protocol: Certificate Verify

Handshake Type: Certificate Verify (15)

Length: 880

```

0000 08 00 00 10 00 0e 00 10 00 0a 00 08 07 68 33 71  ..
0010 78 2d 30 31 0b 00 02 fa 00 00 02 f6 00 02 f1 30  x-
0020 82 02 ed 30 82 01 d5 02 14 3b 86 06 19 77 97 96  ..
0030 7e f9 50 8b 92 f6 9a bf 8c ad 78 1a 27 30 0d 06  ~.
0040 09 2a 86 48 86 f7 0d 01 01 0b 05 00 30 45 31 0b  .*
0050 30 09 06 03 55 04 06 13 02 41 55 31 13 30 11 06  0.
0060 03 55 04 08 0c 0a 53 6f 6d 65 2d 53 74 61 74 65  .U
0070 31 21 30 1f 06 03 55 04 0a 0c 18 49 6e 74 65 72  1!
0080 6e 65 74 20 57 69 64 67 69 74 73 20 50 74 79 20  ne
0090 4c 74 64 30 1e 17 0d 32 30 30 33 32 33 31 36 30  Lt
00a0 37 35 34 5a 17 0d 34 37 30 38 30 39 31 36 30 37  75.
00b0 35 34 5a 30 21 31 0b 30 09 06 03 55 04 06 13 02  54.
00c0 47 43 21 12 20 10 06 02 55 04 02 00 00 71 75 60  60.

```

Frame (1307 bytes)

Decrypted TLS (1086 bytes)

capture.pcap						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
Apply a display filter ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	49860 → 4433 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK
2	0.000018608	127.0.0.1	127.0.0.1	TCP	74	4433 → 49860 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS
3	0.000041696	127.0.0.1	127.0.0.1	TCP	66	49860 → 4433 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=34
4	0.000459780	127.0.0.1	127.0.0.1	TLSv1...	583	Client Hello (SNI=localhost)
5	0.000473296	127.0.0.1	127.0.0.1	TCP	66	4433 → 49860 [ACK] Seq=1 Ack=518 Win=65024 Len=0 TSval=
6	0.003373793	127.0.0.1	127.0.0.1	TLSv1...	1307	Server Hello, Change Cipher Spec, Encrypted Extensions,
7	0.003398753	127.0.0.1	127.0.0.1	TCP	66	49860 → 4433 [ACK] Seq=518 Ack=1242 Win=72576 Len=0 TSV
8	0.004674882	127.0.0.1	127.0.0.1	TLSv1...	120	Change Cipher Spec, Finished
9	0.006101947	127.0.0.1	127.0.0.1	QMux	565	[1 record], QX_TP
10	0.006119489	127.0.0.1	127.0.0.1	QMux	147	[1 record], QX_TP
11	0.006444846	127.0.0.1	127.0.0.1	HTTP3	157	[1 record]SETTINGS, STREAM(7), STREAM(11), STREAM(15)
12	0.006681138	127.0.0.1	127.0.0.1	HTTP3	256	[1 record]SETTINGS, STREAM(6), STREAM(10), STREAM(0), F
13	0.007300149	127.0.0.1	127.0.0.1	HTTP3	164	[1 record]HEADERS: 404 Not Found, DATA
14	0.007704646	127.0.0.1	127.0.0.1	QMux	96	[1 record], CC
15	0.007989843	127.0.0.1	127.0.0.1	TCP	66	49860 → 4433 [FIN, ACK] Seq=883 Ack=1930 Win=72960 Len=
16	0.008189303	127.0.0.1	127.0.0.1	TCP	66	4433 → 49860 [FIN, ACK] Seq=1930 Ack=884 Win=65536 Len=
17	0.008212213	127.0.0.1	127.0.0.1	TCP	66	49860 → 4433 [ACK] Seq=884 Ack=1931 Win=72960 Len=0 TSV

▶ Frame 9: Packet, 565 bytes on wire (4520 bits), 565 bytes captured (4520 bits) on interface lo, id 0	0000	3a ff 51 53 30 0d 0a 0d 0a 31 01 04 0
▶ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)	0010	04 04 80 98 96 80 05 04 80 0f 42 40 0
▶ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1	0020	42 40 07 04 80 0f 42 40 08 02 40 64 0
▶ Transmission Control Protocol, Src Port: 4433, Dst Port: 49860, Seq: 124, Ack: 582, Len: 499	0030	c5 71 c5 94 29 cd 08 45 02 7f fe 17
▶ Transport Layer Security		
▼ QMux		
Record Size: 58		
▼ QMux Record 0 (Size: 58)		
▼ QX_TRANSPORT_PARAMETERS		
Length: 49		
▶ Parameter: max_idle_timeout (len=4) = 30000 ms		
▶ Parameter: initial_max_data (len=4) = 10000000		
▶ Parameter: initial_max_stream_data_bidi_local (len=4) = 10000000		
▶ Parameter: initial_max_stream_data_bidi_remote (len=4) = 10000000		
▶ Parameter: initial_max_stream_data_uni (len=4) = 10000000		
▶ Parameter: initial_max_streams_bidi (len=2) = 100		
▶ Parameter: initial_max_streams_uni (len=2) = 100		
▼ Parameter: max_record_size (len=2) = 16382		
Parameter ID: max_record_size (0x0571c59429cd0845)		
Parameter Length: 2		
Parameter Value: 7ffe		
max_record_size: 16382		

capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	49860 → 4433 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=3427743191 TSecr=0 WS=
2	0.000018608	127.0.0.1	127.0.0.1	TCP	74	4433 → 49860 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=3427743191
3	0.000041696	127.0.0.1	127.0.0.1	TCP	66	49860 → 4433 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=3427743191 TSecr=3427743191
4	0.000459780	127.0.0.1	127.0.0.1	TLSv1...	583	Client Hello (SNI=localhost)
5	0.000473296	127.0.0.1	127.0.0.1	TCP	66	4433 → 49860 [ACK] Seq=1 Ack=518 Win=65024 Len=0 TSval=3427743191 TSecr=3427743191
6	0.003373793	127.0.0.1	127.0.0.1	TLSv1...	1307	Server Hello, Change Cipher Spec, Encrypted Extensions, Certificate, Certificate Verify,
7	0.003398753	127.0.0.1	127.0.0.1	TCP	66	49860 → 4433 [ACK] Seq=518 Ack=1242 Win=72576 Len=0 TSval=3427743194 TSecr=3427743194
8	0.004674803	127.0.0.1	127.0.0.1	TLSv1...	130	Change Cipher Spec, Finished
9	0.006101947	127.0.0.1	127.0.0.1	QMux	565	[1 record], QX_TP
10	0.006119489	127.0.0.1	127.0.0.1	QMux	147	[1 record], QX_TP
11	0.006119489	127.0.0.1	127.0.0.1	SETTINGS	157	[1 record]SETTINGS, STREAM(7), STREAM(11), STREAM(15)
12	0.006681138	127.0.0.1	127.0.0.1	HTTP3	256	[1 record]SETTINGS, STREAM(6), STREAM(10), STREAM(0), HEADERS: GET https://localhost/index
13	0.007300149	127.0.0.1	127.0.0.1	HTTP3	164	[1 record]HEADERS: 404 Not Found, DATA
14	0.007784646	127.0.0.1	127.0.0.1	QMux	96	[1 record], CC
15	0.007989843	127.0.0.1	127.0.0.1	TCP	66	49860 → 4433 [FIN, ACK] Seq=883 Ack=1930 Win=72960 Len=0 TSval=3427743199 TSecr=342774319
16	0.008189303	127.0.0.1	127.0.0.1	TCP	66	4433 → 49860 [FIN, ACK] Seq=1930 Ack=884 Win=65536 Len=0 TSval=3427743199 TSecr=342774319
17	0.008212213	127.0.0.1	127.0.0.1	TCP	66	49860 → 4433 [ACK] Seq=884 Ack=1931 Win=72960 Len=0 TSval=3427743199 TSecr=3427743199

Frame 12: Packet, 256 bytes on wire (2048 bits), 256 bytes captured (2048 bits) on interface lo, id 0

Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)

Transmission Control Protocol, Src Port: 49860, Dst Port: 4433, Seq: 663, Ack: 1832, Len: 190

Transport Layer Security

QMux

Record Size: 166

QMux Record 0 (Size: 166)

- STREAM id=2 fin=0 off=0 len=19 dir=Unidirectional origin=Client-initiated
- STREAM id=6 fin=0 off=0 len=1 dir=Unidirectional origin=Client-initiated
- STREAM id=10 fin=0 off=0 len=1 dir=Unidirectional origin=Client-initiated
- STREAM id=0 fin=1 off=0 len=94 dir=Bidirectional origin=Client-initiated
- STREAM id=14 fin=1 off=0 len=26 dir=Unidirectional origin=Client-initiated

Hypertext Transfer Protocol Version 3

- UNI STREAM: Control Stream off=0

Hypertext Transfer Protocol Version 3

- UNI STREAM: QPACK Encoder Stream off=0

Hypertext Transfer Protocol Version 3

- UNI STREAM: QPACK Decoder Stream off=0

Hypertext Transfer Protocol Version 3

- Request Stream
 - Reserved (GREASE) len=0
 - Reserved (GREASE) len=18
- HEADERS len=56
 - [Stream ID: 0]
 - Type: HEADERS (0x0000000000000001)
 - Length: 56
 - Frame Payload: 0000d1d75086a0e41d139d0951886d5485f2bce9a685f509fa2d20d0b76a6df2b476147d56b2d6
 - [Decoded Headers Length: 149]
 - [Headers Count: 5]
 - Header: :method: GET
 - Header: :scheme: https
 - Header: :authority: localhost
 - Header: :path: /index.html
 - Header: user-agent: lucas-qmux-break-rules-and-buy-me-sake/0.1
 - [Full request URI: https://localhost/index.html]

Hypertext Transfer Protocol Version 3

- UNI STREAM: Reserved (GREASE) off=0

Open issues

Only 4 open issues:

- Could QMux optionally leverage the existing QUIC TLS negotiation? [#17](#)
- Require TCP [#41](#) (editorial)
- QMux provided ALPN [#60](#)
- Revisit graceful close? [#67](#)

Which substrates does this document define? [#41](#)

- TLS-over-TCP only?
 - ALPN is present
 - Move Unix domain sockets and HTTP CONNECT to appendix
 - No SVCB for other substrates
- Or any suitable transport?
 - Perhaps only define ALPN for TLS-over-TCP?

Could QMUX optionally leverage the existing QUIC TLS negotiation? [#17](#)

- Today, transport provides security properties
- Should QMux have a mode where it provides QUIC CRYPTO frames?
 - Useful for insecure substrates (e.g. RDMA)
 - Requires CRYPTO frames + per-record encryption => rebuilds a QUIC/TLS stack
- Or should insecure substrates add TLS
 - If a substrate can't add TLS messages, why can it add QUIC messages?

QMux provided ALPN [#60](#)

- If not running over TLS, MoQ and others still need to negotiate the application protocol

Options:

- Negotiate it out-of-band
- Use distinct ports for different protocols
- Provide a QX_TRANSPORT_PARAMETERS or QX_ALPN(S)
 - Single protocol, fail if not matching *or*
 - Real ALPN negotiation with multiple offered protocols

Graceful close [#67](#)

- Requiring FIN / “graceful close” for immediate closure is probably useful
- Recent conversations about “silent close” for TCP on idle timeout
- Proposal:

For protocols where silent data loss is possible, implementations SHOULD perform graceful shutdown on idle, but for protocols where this is not a concern, or where this race will elicit a RST anyway, implementations SHOULD perform a silent idle shutdown.

Questions / Next Steps